

PORTFOLIO PROJECT • MULTI-LEVEL

Northwind Analytics Project

A Level-by-Level SQL Server Project — From Setup to a Professional GitHub Case Study

Database: Northwind Traders | Engine: Microsoft SQL Server | 7 Levels + Setup

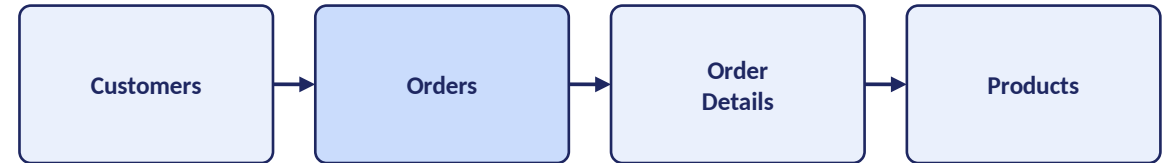
What You Are Building, and Why It Matters

Your Goal

Work through the Northwind Traders database level by level — from your first SELECT to real analytics case studies — and turn it into a project you can show in interviews.

Ground Rules

- Everything runs on Microsoft SQL Server.
- Each level is a set of questions — don't move to the next level until you can answer all of them with working SQL.
- Levels get harder as you go. Level 7 should read like real workplace analysis, not a classroom exercise.
- Save every script — together they become your GitHub repository at the end.



Products also connect to Categories and Suppliers. Orders also connect to Employees and Shippers.

The 8 Sections of This Project

Setup · L1 Foundations · L2 Joins · L3 Aggregation · L4 Subqueries & CTEs · L5 Window Functions · L6 Views & Procedures · L7 Business Case Studies

Final Deliverable

A public GitHub repository with your SQL scripts for every level, your ER diagram, your schema report, and a README that reads like a real analytics case study.

1

FOUNDATION • BEFORE LEVEL 1

Setup — Install, Diagram, and Document the Database

Questions To Answer

- Do you have SQL Server (Developer or Express edition, free) installed and running, with SSMS or Azure Data Studio connected to it?
- Have you restored the Northwind database on your own server, and can you run `SELECT * FROM Customers;` without errors?
- Can you name every table in Northwind, and explain in one sentence what each one stores?

DELIVERABLE A — SETUP

A short note confirming your SQL Server edition, how you restored Northwind, and one issue you solved along the way.

DELIVERABLE B — ER DIAGRAM

A full Entity-Relationship diagram of Northwind — every table, every Primary Key, every Foreign Key — built with SSMS Database Diagrams, dbdiagram.io, or draw.io.

DELIVERABLE C — SCHEMA REPORT

A written report (1–2 pages) listing every table, its columns, and how it relates to the other tables. This is your reference document for every level that follows.

Do not start Level 1 until all three deliverables are done — they are what make every later query easy to write.



Foundations — Filtering and Sorting

Questions To Answer

- Which products currently have zero units in stock?
- List all customers based in a specific country, sorted alphabetically by company name.
- What are the 10 most expensive products in the catalog?
- Which product categories exist, listed without duplicates?

SKILLS THIS LEVEL BUILDS

- SELECT & column selection
- WHERE filtering
- ORDER BY
- TOP / DISTINCT

DELIVERABLE

Save your answers as 01-foundations.sql, with a one-line comment above each query stating the question it answers.

Joins — Connecting the Tables

Questions To Answer

- For every order, show the customer's company name and the employee who processed it.
- For every product, show its category name and its supplier's name in one row.
- Which customers have never placed a single order?
- For every order, show which shipping company delivered it.

SKILLS THIS LEVEL BUILDS

- INNER JOIN
- LEFT JOIN
- Joining 3+ tables
- Filtering unmatched rows

DELIVERABLE

Save your answers as 02-joins.sql. Try writing the customers-with-no-orders question two different ways.

Aggregation — Summarizing the Data

Questions To Answer

- What total revenue has each product category generated?
- How many orders has each employee handled?
- What is the average order value per customer?
- Which categories contain more than 10 products?

SKILLS THIS LEVEL BUILDS

- COUNT, SUM, AVG
- GROUP BY
- HAVING vs. WHERE
- Rounding and formatting results

DELIVERABLE

Save your answers as 03-aggregations.sql. Make sure every total is rounded to a sensible number of decimals.

Subqueries & CTEs — Queries Inside Queries

Questions To Answer

- Which customers spend more than the average customer, across all customers?
- What are the 3 best-selling products overall, by total revenue?
- Using a CTE, what is the best-selling product within each category?
- Which products are priced above the average price within their own category?

SKILLS THIS LEVEL BUILDS

- Scalar & correlated subqueries
- WITH (CTEs)
- Comparing rows to group averages
- Readability vs. nested subqueries

DELIVERABLE

Save your answers as 04-subqueries-cte.sql. For at least one query, write it both as a subquery and as a CTE, and compare.

Window Functions — Ranking and Trends

Questions To Answer

- Rank all customers by total spending, from highest to lowest.
- Calculate a running total of monthly revenue across the full order history.
- For each month, how does revenue compare to the previous month?
- What are the top 2 products by revenue within each category?

SKILLS THIS LEVEL BUILDS

- RANK / DENSE_RANK / ROW_NUMBER
- PARTITION BY
- Running totals
- LAG / LEAD

DELIVERABLE

Save your answers as 05-window-functions.sql. This level is what separates a beginner script from an analyst's script.



Reusable Building Blocks

Questions To Answer

- Create a view that shows every order line with the customer name and product name already joined in.
- Create a stored procedure that returns all orders for a given CustomerID, passed in as a parameter.
- Create a stored procedure that accepts a start and end date and returns total revenue for that range.
- Which column would benefit most from an index if this database had millions of orders, and why?

SKILLS THIS LEVEL BUILDS

- CREATE VIEW
- CREATE PROCEDURE with parameters
- Basic indexing concepts
- Writing reusable SQL

DELIVERABLE

Save your answers as 06-views-procedures.sql, with a short comment explaining what each object is for.

Business Case Studies — Think Like an Analyst

- Who are the top 10 customers by lifetime revenue, and what share of total revenue do they represent?
- What is the month-over-month revenue growth trend for the most recent full year of data?
- Which employees are the top performers by revenue, and does their strength vary by product category?
- Which customers haven't ordered in the last 6 months, and should be flagged as at risk of churning?
- Which country generates the highest average order value, and is that a market worth investing more in?
- If the company had to discontinue three products based on the data alone, which would you recommend, and why?

DELIVERABLE

Save your answers as 07-business-case-studies.sql. For every question, add 2–4 sentences of plain-language insight — this is the file recruiters will actually read.

THIS IS THE SLIDE RECRUITERS READ

Anyone can write a SELECT. Answering questions like these — and explaining the business reasoning behind each result in plain language — is what makes this a professional project instead of a school exercise.

Package It — Build Your GitHub Portfolio Repo

Your README Must Answer

- What is Northwind, and what tools did you use (SQL Server edition, SSMS/Azure Data Studio)?
- What does the schema look like — link to your ER diagram and schema report from Step 1.
- What did each level teach you, in one line per level?
- What are your 2–3 strongest business insights from Level 7, front and center?

SUGGESTED REPO STRUCTURE

```
northwind-sql-server-project/  
├── README.md  
├── diagrams/  
│   └── er-diagram.png  
├── reports/  
│   └── schema-report.md  
└── queries/  
    ├── 01-foundations.sql  
    ├── 02-joins.sql  
    ├── 03-aggregations.sql  
    ├── 04-subqueries-cte.sql  
    ├── 05-window-functions.sql  
    ├── 06-views-procedures.sql  
    └── 07-business-case-studies.sql
```

BEFORE YOU PUBLISH

Final Checklist

- ☐ SQL Server + Northwind installed and running locally
- ☐ ER diagram covers every table, PK, and FK
- ☐ Schema report explains every table in plain language
- ☐ Level 1–3 scripts run cleanly with correct results
- ☐ Level 4–5 scripts use at least one CTE and one window function
- ☐ Level 6 includes a working view and a parameterized procedure
- ☐ Level 7 answers include written business insight, not just SQL
- ☐ README explains the project end-to-end in under two minutes
- ☐ Repo folders match the suggested structure

RESOURCES

Everything You Need, Linked

Northwind for SQL Server (official Microsoft sample)

github.com/microsoft/sql-server-samples/tree/master/samples/databases/northwind-pubs

SQL Server Downloads (Developer/Express, free)

microsoft.com/en-us/sql-server/sql-server-downloads

SQL Server Management Studio (SSMS)

learn.microsoft.com/sql/ssms/download-sql-server-management-studio-ssms

dbdiagram.io — draw your ER diagram online

dbdiagram.io

GitHub Docs — About READMEs

docs.github.com/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/about-readmes